

Lec 17.

Recap

① Total law of expectation

$$E[X] = E[E[X|Y]].$$

② Total law of Variance

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y]).$$

③ Independence of 2 r.v.s
" " n r.v.s.

$$P_{X,Y}(x,y) = P_X(x)P_Y(y) \\ \Leftrightarrow P_{X|Y}(x) = P_X(x).$$

$$P_{X_1, \dots, X_n}(x_1, \dots, x_n) = P_{X_1}(x_1)P_{X_2}(x_2) \dots P_{X_n}(x_n).$$

$$\rightarrow E[g(X)h(Y)] = E[g(X)]E[h(Y)]$$

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y).$$

$$\text{Var}(X_1+X_2+\dots+X_n) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n).$$

Logistics

Quiz on Monday 6-7pm.

Outline.pdf.

Programming Assignment.
(before Spring break)

X and Y are said to be uncorrelated

if $\text{Cov}(X, Y) = 0$

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY] - E[X]E[Y].\end{aligned}$$

Independence $\Rightarrow \text{Cov}(X, Y) = 0$

& X, Y

ie, X, Y are

not correlated.

~~⊗~~

$$X = \begin{cases} 1 & \text{w.p. } \frac{1}{3} \\ -1 & \text{w.p. } \frac{1}{3} \\ 0 & \text{w.p. } \frac{1}{3} \end{cases}$$

$$Y = \begin{cases} 1 & \text{if } X=0 \\ 0 & \text{otherwise} \end{cases}$$

$$E[XY] = 0$$

$$E[X] = \frac{1}{3} \times 1 + \frac{1}{3} \times -1 + \frac{1}{3} \times 0 = 0$$

$$\Rightarrow E[XY] = E[X]E[Y]$$

$$P_{X|Y=0}(x) = \begin{cases} 0 & x=0 \\ ? & x=-1 \\ ? & x=1 \end{cases}$$

$$\neq P_X(x)$$

$\Rightarrow X, Y$ are not independent

For any X, Y

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Correlation of R.V.s X, Y

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}$$

Lemma:

$$-1 \leq \rho(X, Y) \leq 1 \Leftrightarrow \rho^2(X, Y) \leq 1$$

$$(\text{Cov}(X, Y))^2 \leq \text{Var}(X) \text{Var}(Y)$$

Cauchy-Schwartz inequality

$$E[(\bar{X} - \alpha \bar{Y})^2] \geq 0 \quad \forall \alpha \in \mathbb{R}$$

$$= E[\bar{X}^2 + \alpha^2 \bar{Y}^2 - 2\alpha \bar{X}\bar{Y}]$$

$$= E[\bar{X}^2] + \alpha^2 E[\bar{Y}^2] - 2\alpha E[\bar{X}\bar{Y}]$$

$$= \text{Var}(\bar{X}) + \alpha^2 \text{Var}(\bar{Y}) - 2\alpha \text{Cov}(\bar{X}, \bar{Y}) \geq 0$$

For above polynomial to be $\geq 0 \quad \forall \alpha \in \mathbb{R}$,

$$[2 \text{Cov}(\bar{X}, \bar{Y})]^2 - 4 \text{Var}(\bar{X}) \text{Var}(\bar{Y}) \leq 0$$

$$\Rightarrow \text{Cov}(\bar{X}, \bar{Y})^2 \leq \text{Var}(\bar{X}) \text{Var}(\bar{Y})$$

$$\Rightarrow \rho(\bar{X}, \bar{Y})^2 \leq 1$$

When is equality satisfied?

iff $\bar{Y}$ is a linear function of $\bar{X}$.

ie, $\bar{Y} = c\bar{X}$ w.p 1.

Logistics

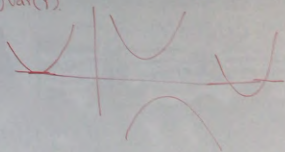
Quiz on Monday 6-7pm

Outline.pdf

Programming Assignment
(before Spring break)

$$a\alpha^2 + b\alpha + c = 0$$

$$\alpha = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$I(X; Y) = \sum_{\substack{x \in \mathcal{X} \\ y \in \mathcal{Y}}} P_{X,Y}(x,y) \log \frac{P_{X,Y}(x,y)}{P_X(x)P_Y(y)}$$

$$I(X; Y) = 0 \iff X, Y \text{ are independent.}$$

Sum of two independent RVs.

$$\textcircled{1} \quad X \sim \text{Poisson}(\lambda_1), \quad Y \sim \text{Poisson}(\lambda_2).$$

$$P_X(x) = \frac{e^{-\lambda_1} \lambda_1^x}{x!}, \quad P_Y(y) = \frac{e^{-\lambda_2} \lambda_2^y}{y!}$$

$x=0,1,2,\dots$

$$Z = X + Y, \quad X \text{ and } Y \text{ are independent}$$

$$P_Z(z) = \sum_{x,y: x+y=z} P_{X,Y}(x,y)$$

Proof for $I(X; Y) = 0 \iff X, Y \text{ are independent}$

(a) $\iff$ is clear from defn.

(b) $\Rightarrow$

We'll show $I(X; Y) \geq 0$ first

$$I(x; y) = \sum_{x, y} P_{x, y}(x, y) \log \frac{P_{x, y}(x, y)}{P_x(x) P_y(y)}$$

Using
 $\log x \leq 1 - x$.

$$\leftarrow = - \sum_{x, y} P_{x, y}(x, y) \log \frac{P_x(x) P_y(y)}{P_{x, y}(x, y)}$$

$$\Rightarrow -\log x \geq x - 1$$

$$\geq - \sum_{x, y} P_{x, y}(x, y) \left[\frac{P_x(x) P_y(y)}{P_{x, y}(x, y)} - 1 \right]$$

$$= - \sum_{x, y} P_x(x) P_y(y) + \sum_{x, y} P_{x, y}(x, y)$$

$$= -1 + 1 = 0.$$



Equality holds above iff

$$\log \left(\frac{P_x(x) P_y(y)}{P_{x, y}(x, y)} \right) = 1 - \frac{P_x(x) P_y(y)}{P_{x, y}(x, y)}.$$

iff

$$\frac{P_x(x) P_y(y)}{P_{x, y}(x, y)} = 1 \quad \forall x, y.$$

$$P_Z(z) = \sum_{\substack{x=0 \\ y=z-x}}^z P_X(x) P_Y(y)$$

$$= \sum_{\substack{x=0 \\ y=z-x}}^z \left[\frac{e^{-\lambda_1} \lambda_1^x}{x!} \right] \frac{e^{-\lambda_2} \lambda_2^{z-x}}{(z-x)!}$$

$$= \sum_{x=0}^z e^{-(\lambda_1 + \lambda_2)} \frac{\lambda_1^x \lambda_2^{z-x}}{x! (z-x)!} \binom{z}{x}$$

$$= \frac{e^{-(\lambda_1 + \lambda_2)}}{z!} \sum_{x=0}^z \lambda_1^x \lambda_2^{z-x} \binom{z}{x}$$

$$= \frac{e^{-(\lambda_1 + \lambda_2)}}{z!} (\lambda_1 + \lambda_2)^z$$

$$Z \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

② Total Law of expectation example

(To find $E[X]$, $\text{Var}(X)$)

Geometric RV

for $X \sim \text{Geometric}(p)$

Memoryless property

↗ LHS

$$\boxed{P(X > t + 1 \mid X > t)} = P(X > t)$$

$$P(X > t) = \sum_{x=t+1}^{\infty} P(X=x)$$

$$= \sum_{x=t+1}^{\infty} (1-p)^{x-1} p$$

$$\frac{1}{1-(1-p)} = \frac{1}{p}$$

$$= (1-p)^t \cancel{p} \left[1 + \cancel{(1-p)} + \dots \right]$$

$$P(X > t) = (1-p)^t$$

$$P(v \rightarrow t + \tau | X > t) = \frac{P(X > t + \tau, X > t)}{P(X > t)}$$

$$= \frac{P(X > t + \tau)}{P(X > t)}$$

$$= \frac{(1-p)^{t+\tau}}{(1-p)^t} = (1-p)^\tau = P(X > \tau)$$

$$A = T_1: \{X > 1\} \quad A^c: H_1 = \{X = 1\}$$

$$E[X] = P(T_1) E[X | T_1] + P(H_1) E[X | H_1]$$

$$\begin{aligned} & X | H_1 = 1 \text{ w.p. } 1 \\ & P(X=x | H_1) = \begin{cases} P(X=1) & x=1 \\ 0 & x \neq 1 \end{cases} \Rightarrow E[X | H_1] = 1 \end{aligned}$$

$$E[X] = (1-p) \underbrace{E[X|T_1]} + p \cdot 1$$

$$E[X|T_1] = 1 + E[X] \quad (\text{claim})$$

Experiment restarts after the 1st unsuccessful toss.

$$E[X|T_1] = \sum_{x=1}^{\infty} x P_{X|T_1}(x)$$

$$P_{X|T_1}(x) = P(x) \mid X \geq 1$$

$$= \frac{P(\{X=x\} \cap \{X \geq 1\})}{P(\{X \geq 1\})}$$

$$= \begin{cases} 0 & x=1 \\ \frac{P_X(x)}{P(X \geq 1)} & x > 1 \end{cases} = \begin{cases} 0 & x=1 \\ \frac{(1-p)^{x-1} p}{(1-p)} & x > 1 \end{cases}$$

$$P_{X|T_1}(x) = \begin{cases} (1-p)^{x-2} p & x > 1 \text{ or } x \geq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$E[X|T_1] = \sum_{x=2}^{\infty} x (1-p)^{x-2} p$$

$$\begin{aligned} x = \hat{x} + 1 &= \sum_{\hat{x}=1}^{\infty} (\hat{x} + 1) (1-p)^{\hat{x}-1} p \\ &= \sum_{\hat{x}=1}^{\infty} \hat{x} (1-p)^{\hat{x}-1} p + \sum_{\hat{x}=1}^{\infty} (1-p)^{\hat{x}-1} p \\ &= E[X] + 1 \end{aligned}$$

$$E[X] = (1-p) [1 + E[X]] + p$$

$$\Rightarrow E[X] [1 - (1-p)] = 1-p+p = 1$$

$$\Rightarrow \boxed{E[X] = \frac{1}{p}}$$

$$E[X^2] = E[X^2|T_1] (1-p) + \underbrace{E[X^2|H_1]}_{=1} p$$

$$= E[X^2|T_1] (1-p) + p$$

$$E[X^2|T_1] = E[(1+X)^2]$$

Key fact: $X|T_1 = 1+X$ follows from $P_{X|T_1}(x) = P_X(x-1)$